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MOVING BODY CONTROL SYSTEM, CONTROL METHOD THEREOF, MOVING BODY, AND STORAGE MEDIUM

Abstract

A moving body control system generates a reference route indicating a route from a position of the moving body on which the moving body travels and determines a part of the reference route to be referred to for controlling travel of the moving body out of the reference route. The system determines a control amount for controlling travel of the moving body with reference to the determined part of the reference route. The system determines a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of International Patent Application No. PCT/JP2023/043636 filed on Dec. 6, 2023, which claims priority to and the benefit of Japanese Patent Application No. 2022-195843 filed on Dec. 7, 2022, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a moving body control system, a control method thereof, a moving body, and a storage medium.

BACKGROUND ART

[0003] Recently, there is an increasing demand for ultra-compact moving bodies (micro mobility vehicles) for supporting movements of people in small regions. Micro mobility vehicles include vehicles each having a riding capacity of one person or so, vehicles that travel carrying a baggage along with a person, instead of carrying the person and the like. The micro mobility vehicles require an autonomous moving technology in order to be able to travel in both moving regions of automobiles and moving regions of pedestrians.

[0004] In the autonomous moving technology, there is known a technology of generating a travel route for a moving body to travel and controlling travel of the moving body along the generated route. Regarding generation of the travel route, there is known a technology of generating a travel locus of a preceding vehicle by applying a straight line or a curved line to an observation point indicating a position where the preceding vehicle has passed in a device that controls travel of a self-vehicle so as to follow the preceding vehicle traveling in front of the self-vehicle (Patent Literature 1).

CITATION LIST

Patent Literature

[0005] PTL 1: Japanese Patent Laid-Open No. 2020-32844

SUMMARY OF INVENTION

Technical Problem

[0006] Note that, in order to avoid a right or left turn at an intersection and an obstacle, a micro mobility vehicle is required to achieve both highly responsive control for accurately following a route to travel and control for stabilizing a behavior of a moving body in consideration of a passenger.

[0007] The present invention has been achieved in view of the above-described problem, and an object thereof is to implement a technique capable of achieving both responsiveness to a route to travel and stability of a behavior.

Solution to Problem

[0008] According to the present invention, a moving body control system that controls an operation of a moving body, the moving body control system comprising: one or more processors; and a

memory storing instructions which, when the instructions are executed by the one or more processors, cause the moving body control system to perform: generating a reference route indicating a route from a position of the moving body on which the moving body travels; determining a part of the reference route to be referred to for controlling travel of the moving body out of the reference route; and determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, wherein determining the part of the reference route includes determining a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.

Advantageous Effects of Invention

[0009] According to the present invention, it is possible to achieve both responsiveness to a route to travel and stability of a behavior.

[0010] Other features and advantages of the present invention will be apparent from the following description taken in conjunction with the accompanying drawings. Note that the same reference numerals denote the same or like components throughout the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and, together with the description, serve to explain principles of the invention.

[0012] FIG. 1A is a diagram (1) illustrating a configuration example of a moving body according to an embodiment of the present invention.

[0013] FIG. 1B is a diagram (2) illustrating a configuration example of a moving body according to an embodiment of the present invention.

[0014] FIG. 2 is a block diagram illustrating a configuration example of a control system of the moving body according to the embodiment.

[0015] FIG. 3 is a block diagram illustrating a functional configuration example related to a control unit of the moving body according to the embodiment.

[0016] FIG. 4 is a diagram for explaining an outline of travel control at an intersection according to the embodiment.

[0017] FIG. 5 is a flowchart illustrating a series of operations of travel control processing in the moving body according to the embodiment.

[0018] FIG. 6 is a diagram for explaining determination of a reference route length using a curvature change point according to the embodiment.

[0019] FIG. 7 is a diagram for explaining a first example regarding determination of a reference route length using a terminal point of a straight road according to the embodiment.

[0020] FIG. 8 is a diagram for explaining a second example regarding determination of a reference route length using a terminal point of a straight road according to the embodiment.

[0021] FIG. 9 is a flowchart illustrating a series of operations of reference route length determination processing according to the embodiment.

DESCRIPTION OF EMBODIMENTS

[0022] Hereinafter, an embodiment will be described in detail with reference to the attached drawings. Note that, the following embodiment does not limit the invention recited in claims, and not all combinations of features described in the embodiment are essential to the invention. Two or more features of a plurality of the features described in the embodiments may be optionally

combined. The same reference numeral is given to the same or similar configurations, and the description will be repeated.

[0023] In the following embodiment, as an example of a moving body, which is a micro mobility vehicle, an ultra-compact electric vehicle having a riding capacity of one person or so will be described as an example. Note that, micro mobility vehicles may include a vehicle that travels carrying a baggage along with a person, instead of carrying the person. The present embodiment is not limited to an example in which the moving body is the electric vehicle, and is applicable to any moving body other than the electric vehicle. Furthermore, in the following description, a moving body including one driven wheel will be described as an example, but the driven wheel is not necessarily provided, and the number of driven wheels is not limited to one and may be two or more.

[0024] In the above-described moving body such as the micro mobility vehicle, it is useful when autonomous travel is implemented in consideration of riding of a person, frequent changes in target position, and non-use of a high-precision map. The micro mobility vehicle does not always travel on a specific fixed route, and it is necessary to appropriately travel in a region in which a high-precision map is not prepared in order to be able to travel in both moving regions of vehicles and moving regions of pedestrians. Furthermore, for example, as in a case of traveling in a shopping mall or an event venue, there is a need to travel while appropriately avoiding a plurality of obstacles in a situation in which the obstacles are irregularly present. In a case where a person rides on the micro mobility vehicle, travel with high stability in consideration of ride comfort is required.

[0025] A moving body **100** according to the present embodiment recognizes a travel region and autonomously travels along a generated route without using a high-precision map. Since the autonomous travel is performed without using a high-precision map, a region in which the moving body **100** can travel is identified using information recognized from an output of a detection unit to be described later. For example, the moving body **100** identifies a road structure such as a straight road and an intersection on a road, and generates a route (reference route) for traveling on the identified straight road and intersection on the road. The moving body **100** determines a length of the route to be actually referred to out of the reference route, and determines a control amount of the moving body **100** with reference to the reference route in a range of the determined length. The moving body **100** controls a drive system according to the determined control amount. Note that, in the present embodiment, a route on a straight road and an intersection will be described as an example, but other routes that require steering including right and left turns, start, and stop of the moving body may be included. For example, the present invention can also be applied to a case of changing lanes, moving between a sidewalk and a roadway, a case of traveling while avoiding obstacles, a case of traveling on a road in which a straight road and a curved road are combined and the like. An intersection is a road structure in which two roads are connected, and may include an intersection of another form such as a T-junction in addition to a crossroads. An intersection may include an L-shaped travel path at which two roads are connected.

Configuration of Moving Body

[0026] A configuration of the moving body **100** will be described with reference to FIG. 1. FIG. 1A illustrates a side view of the moving body **100** according to the present embodiment, and FIG. 1B illustrates an internal configuration of the moving body **100**. In the drawings, arrow X indicates a front-and-rear direction of the moving body **100**, F indicates the front, and R indicates the rear. Arrows Y and Z indicate a width direction (left-and-right direction) and an up-and-down direction of the moving body **100**, respectively.

[0027] The moving body **100** is an electric autonomous vehicle provided with a travel unit **112** and using a battery **113** as a main power supply. The battery **113** is, for example, a secondary battery such as a lithium ion battery, and the moving body **100** autonomously travels by the travel unit **112** with electric power supplied from the battery **113**. The travel unit **112** has a form of a three-wheeled vehicle provided with a pair of left and right drive wheels **120**, which are front wheels,

and one driven wheel **121**, which is a rear wheel. Note that, the travel unit **112** may have another form such as a form of a four-wheeled vehicle. The moving body **100** is provided with, for example, a seat for one person **111**.

[0028] The travel unit **112** is provided with a drive mechanism **122**. The drive mechanism **122** is a mechanism that rotates a corresponding drive wheel **120** with motors **122a** and **122b** as drive sources. By rotating each of the drive wheels **120**, the drive mechanism **122** can move the moving body **100** forward or backward. The drive mechanism **122** can also change an advancing direction of the moving body **100** by generating a difference in rotation between the motors **122a** and **122b**. The travel unit **112** is provided with the driven wheel **121**. The driven wheel can make a turn with a Z direction as a rotation axis.

[0029] The moving body **100** is provided with detection units **114** to **116**, each of which detects a target object around the moving body **100**. The detection units **114** to **116** form an external sensor group that monitors the periphery of the moving body **100**. In a case of the present embodiment, each of the detection units **114** to **116** is an imaging device that captures an image around the moving body **100**, and is provided with, for example, an optical system such as a lens and an image sensor. However, instead of or in addition to the imaging device, a radar or light detection and ranging (LiDAR) can also be used.

[0030] For example, two detection units **114** are arranged in a front portion of the moving body **100** to be spaced apart from each other in a Y direction, and are mainly used to detect a target object in front of the moving body **100**. The detection units **115** are arranged in a left portion and a right portion of the moving body **100**, and are mainly used to detect target objects on the sides of the moving body **100**. The detection unit **116** is arranged in a rear portion of the moving body **100**, and is mainly used to detect a target object behind the moving body **100**.

[0031] FIG. 2 is a block diagram of a control system of the moving body **100**. The moving body **100** is provided with a control unit (ECU) **130**. The control unit **130** includes one or more processors represented by a CPU, a memory device such as a semiconductor memory, an interface with an external device and the like. The memory device stores a program to be executed by the processor, data to be used by the processor for processing and the like. A plurality of sets of the processor, memory device, and interface may be provided for respective functions of the moving body **100** so as to be able to communicate with each other.

[0032] The control unit **130** acquires outputs (for example, image information) from the detection units **114** to **116**, input information into an operation unit **131**, voice information input from a voice input device **133** and the like, and executes processing corresponding to each piece of the information. The control unit **130** performs, for example, control of the motors **122a** and **122b** (travel control of the travel unit **112**) and display control of a display panel included in the operation unit **131**, gives a notification to an occupant of the moving body **100** by voice, and outputs information. The control unit **130** may execute processing using a machine learning model for image recognition (for example, deep neural network) on the outputs (for example, image information) from the detection units **114** to **116**. The control unit **130** may execute processing using a machine learning model for voice recognition (for example, deep neural network) on the output (for example, voice information) from the voice input device **133**.

[0033] The voice input device **133** includes, for example, a microphone, and collects a voice of the occupant of the moving body **100**. The control unit **130** can recognize the input voice and execute corresponding processing. A global navigation satellite system (GNSS) sensor **134** receives a GNSS signal, and detects a current position of the moving body **100**.

[0034] A storage device **135** includes a nonvolatile storage medium that stores various pieces of data. The storage device **135** may also store a program to be executed by the processor, data to be used by the processor for processing and the like. The storage device **135** may store various parameters (for example, trained parameters of the deep neural network, hyperparameters and the like) of the machine learning model for voice recognition or image recognition to be executed by

the control unit **130**.

[0035] A communication device **136** is a communication device capable of communicating with an external device (for example, a communication terminal **140** owned by a user) via wireless communication, such as Wi-Fi or 5th generation mobile communication.

[0036] Next, a functional configuration example regarding the control unit **130** will be described with reference to FIG. **3**. A user instruction acquisition unit **301** acquires a user instruction input via the operation unit **131** or the voice input device **133**. The user instruction includes designation of a final target position at which the moving body **100** should arrive. The final target position may be a position of a target object designated by an utterance voice among target objects recognized in the images output by the detection units **114** to **116**. The user instruction may include a change instruction for travel, such as a right turn or a left turn, while the moving body **100** is traveling.

[0037] An image information processing unit **302** recognizes a position, a shape, a travel region and the like of an obstacle on the basis of the outputs (for example, image information) of the detection units **114** to **116**. The position, shape, and travel region of the obstacle, the road structure and the like in front of the moving body **100** are recognized by, for example, applying the machine learning model for image recognition trained in advance to the images obtained from the two detection units **114**. Estimation of a depth from the moving body **100** with the images obtained from the two detection units **114** used as a stereo image may be included. Furthermore, in order to recognize the target object in the image, the machine learning model (for example, deep neural network) for image recognition trained in advance may be used for a monocular image or a stereo image.

[0038] A route generation unit **303** generates a reference route on the basis of the travel region recognized by the image information processing unit **302**. The reference route indicates a route on which the moving body **100** travels. For example, in a case where there is an intersection the road structure of which is a crossroads, and the moving body **100** turns right at the intersection, the route generation unit **303** generates the reference route so as to pass through a position away from the center of the intersection by a first predetermined distance and travel at a position away from a left end of the road after turning right by a second predetermined distance. The route generation unit **303** may further generate the reference route on the basis of an identified obstacle. Note that, in the present embodiment, the route generation unit **303** may generate the reference route by any method, and may generate the reference route by using a known method.

[0039] The control amount determination unit **304** determines a length (reference route length) of the route to be actually referred to out of the reference routes, and determines a control amount of the moving body with reference to the reference route in a range of the determined length. Reference route length determination processing for determining the reference route length will be described later in detail.

[0040] The travel control unit **305** controls the travel of the moving body **100** (for example, controls the motors **122a** and **122b**) according to the control determined by the route generation unit **303**.

Travel Control of Moving Body Using Reference Route

[0041] An outline of travel control of the moving body using the reference route will be described with reference to FIG. **4**. FIG. **4** illustrates a case where the moving body **100** traveling in the X direction turns left at the intersection and then goes straight in the Y direction. A travel path **404** includes, for example, two straight roads and an intersection. Reference sign **407** represents a reference route generated by the route generation unit **303**. On the reference route **407**, a plurality of points (reference points) representing positions on the reference route **407** such as reference points **405** and **406** are schematically illustrated. The reference route **407** includes a straight line portion **401** and a straight line portion **403** for traveling on a straight road, and a curved line portion **402**. The reference point **405** indicates a terminal end of the straight line portion **401** (a start end of the curved line portion **402**), and the reference point **406** indicates a start end of the straight line

portion **403** (a terminal end of the curved line portion **402**).

[0042] In the travel control of the moving body **100**, a travel behavior of the moving body **100** might differ depending on a length to be referred to out of the generated reference route. For example, a case is conceivable where a current position of the moving body **100** is a position of the reference point **408**, and only three meters (for example, three reference points from the reference point **408** to the reference point **405**) out of the reference route **407** is referred to. In this case, it is possible to implement the travel control of quickly responding to a change of the reference route **407** and travel on a locus close to the route of the reference route **407**. On the other hand, in order to quickly respond to the change of the reference route **407**, the moving body **100** easily swings left and right, and the travel behavior might become unstable. In contrast, in a case where 13 meters (for example, 13 reference points from the reference point **408** to the reference point **409**) out of the reference route **407** is referred to, a position of the route after turning left, such as the reference point **409**, is taken into consideration to recent travel control of the moving body. In this case, the travel behavior can be stabilized by taking into consideration the positions of more reference points, but in the example illustrated in FIG. **4**, a degree of traveling on an inner side the reference route **407** is large (a degree of inward turning is large). That is, followability with respect to the reference route **407** decreases as compared with a case of referring to the reference route in a shorter range.

[0043] In the present embodiment, by performing the reference route length determination processing, the length of the reference route to be referred to is made different according to a situation. In particular, the length of the reference route to be referred to is set on the basis of a position of a change point of the travel region. For example, in a case where the moving body **100** travels in the straight line portion **401**, the control amount determination unit **304** determines a length from the moving body **100** to the terminal end of the straight line portion **401** (reference point **405**) as the length of the route to be actually referred to. In a case where the moving body **100** travels in the curved line portion **402**, the control amount determination unit **304** determines a length from the moving body **100** to the terminal end of the curved line portion **402** as the length of the route to be actually referred to. Furthermore, in a case where the moving body **100** travels in the straight line portion **403**, the control amount determination unit **304** determines a length from the moving body **100** to the terminal end of the straight line portion **403** (not illustrated) as the length of the route to be actually referred to. Note that, although the curved line portion **402** illustrated in FIG. **4** is described taking a case where there is one curved line that turns left into consideration, there might be a travel region where curved lines are continuous such that a curved line that turns right and a curved line that turns left are combined as an S-shaped curve. In such curved line, positive/negative of a curvature changes at a joint between the two curved lines. Therefore, a point at which positive/negative of the curvature changes of the curved line is referred to as a curvature change point (also considered as a terminal end of each curved line portion) for convenience.

[0044] Note that, in a case where the reference route is actually generated, there is a case where a straight road and a curved road are not clear, and there is a case where the straight road and the curved road are not divided as illustrated in FIG. **4**. Therefore, in the reference route length determination processing of the present embodiment, the control amount determination unit **304** performs processing of identifying the terminal end of the straight line forming the reference route and processing of identifying the terminal end of the curved line forming the reference route and the change point of the predetermined curvature in parallel, and determines a position distant from the moving body **100** out of the identified positions as the length of the route to be actually referred to.

[0045] In this manner, it is possible to use a longer route that can be identified as a straight line or a curve while referring to a route in a specific type of route such as a straight line or a curved line. As a result, for example, a stable travel behavior can be obtained while reducing the degree of inward turning in the example illustrated in FIG. **4**. That is, it is possible to achieve both responsiveness to

the route to travel and stability of the behavior.

Series of Operations of Travel Control Processing in Moving Body **100**

[0046] Next, a series of operations of travel control processing in the moving body **100** will be described with reference to FIG. 5. Note that, this processing is implemented when the control unit **130** develops and executes the program stored in the storage device **135** on the memory device of the control unit **130**. Note that, a final target position is assumed to be designated in advance by the user.

[0047] At S501, the image information processing unit **302** of the control unit **130** acquires the outputs (image information) of the detection units **114** to **116**. The image information processing unit **302** recognizes a travel region (for example, a road structure), a position, a shape and the like of an obstacle on the basis of the image information by using, for example, a deep neural network.

[0048] At S502, the route generation unit **303** of the control unit **130** generates a reference route on the basis of the recognized road structure. As an example, in a case where the moving body **100** turns right at a crossroads (intersection), the route generation unit **303** generates a reference route so as to pass through a position away from the center of the intersection by a first predetermined distance. The route generation unit **303** further generates the reference route so as to travel at a position away from a left end of the road after turning right by a second predetermined distance. Note that, the reference route generated by the route generation unit **303** is not limited to this example, and may be another route.

[0049] At S503, the control amount determination unit **304** of the control unit **130** executes the above-described reference route length determination processing and determines a length of the route to be actually referred to out of the reference route. A specific operation in the reference route length determination processing will be described later. At S504, the control amount determination unit **304** determines a control amount (velocity v and angular velocity ω) of the moving body **100** according to the reference route in a range of the determined reference route length.

[0050] At S505, the travel control unit **305** of the control unit **130** controls the motors **122a** and **122b** using the control amounts determined by the route generation unit **303** to control the travel of the moving body **100**.

[0051] At S506, the control unit **130** determines whether the final target position is reached. In a case where it is determined that the final target position is not reached, the control unit **130** returns the processing to S501 and repeats the processing. In a case where it is determined that the final target position is reached, the control unit **130** terminates the series of processing. Note that, in this processing, a case where the reference route is generated on the basis of the recognized road structure has been described as an example; however, the reference route can be generated on the basis of not only the road structure but also the travel region recognized on the basis of the image.

Series of Operations of Reference Route Length Determination Processing in Moving Body **100**

[0052] Next, a series of operations of the reference route length determination processing executed by the control amount determination unit **304** will be described with reference to FIG. 9. Note that, this processing is implemented when the control unit **130** develops and executes the program stored in the storage device **135** on the memory device of the control unit **130**. Note that this processing is executed when above-described S503 is started.

[0053] At S901, the control amount determination unit **304** searches for the curvature change point on the reference route and specifies a terminal end position of the curved line. Processing of specifying the terminal end position of the curved line will be described with reference to FIG. 6.

[0054] The example illustrated in FIG. 6 illustrates an example of processing in which the moving body **100** refers to the reference route and determines the reference route length on a road **601** including a road structure that can turn right. Specifically, a reference route **603** is provided, and there is a plurality of reference points on the reference route **603** from a position of a moving body **602**. In the present embodiment, the control amount determination unit **304** selects consecutive reference points (for example, four reference points in a range **604** including a standard reference

point **605**) included in a predetermined range, approximates a circular arc passing through these reference points, and determines a change in curvature of the approximated circular arc. For example, the control amount determination unit **304** sequentially moves the standard reference point away from the position the closest to the moving body **602** to obtain the change in curvature at each reference point. For example, the control amount determination unit **304** obtains an average value and a deviation of curvatures at the selected reference points. In a case where the deviation exceeds a threshold, a position of the standard reference point **605** can be set as the curvature change point. Since the curved line portion in the intersection has a substantially constant curvature, a portion in which the deviation increases, that is, a portion in which the curvature is equal to or smaller than a predetermined value can be the curvature change point. In the example of FIG. **6**, for example, the control amount determination unit **304** can specify the position of the standard reference point **605** as the terminal end position of the curved line.

[0055] At **S902**, the control amount determination unit **304** specifies a terminal end position of a straight line in which the reference route and a straight line from a specific point on the reference route to the moving body have a coincidence degree equal to or greater than a threshold. Processing of specifying the terminal end position of the straight line at this step will be described with reference to FIG. **7**.

[0056] The example illustrated in FIG. **7** illustrates an example of processing in which a moving body **702** that travels on a road **701** refers to a reference route **703** and determines a reference route length. The control amount determination unit **304** performs fitting between a straight line connecting the moving body **702** and the reference point and the reference route sequentially from a reference point distant by a predetermined distance or from an identifiable most distant reference point. For example, the control amount determination unit **304** determines whether the coincidence degree between a straight line connecting the moving body **702** and the reference point **704** and the reference route between the moving body **702** and the reference point **704** is equal to or greater than a threshold. In the determination of the coincidence degree, it may be determined whether deviation from the straight line connecting the moving body **702** and the reference point **704** to the reference route satisfies a certain condition (whether the sum of squares of the deviation is equal to or less than a predetermined threshold). In a case where the straight line connecting the moving body **702** and the reference point **704** and the reference route do not indicate the coincidence degree equal to or greater than the threshold, the control amount determination unit **304** further performs processing on the reference point closer to the moving body **702**. For example, the control amount determination unit **304** can determine the coincidence degree between the straight line connecting the moving body **702** and the reference point **706** and the reference route. In a case where the straight line and the reference route have the coincidence degree equal to or greater than the threshold, the control amount determination unit **304** specifies the reference point **706** as the terminal end position of the straight line. That is, the control amount determination unit **304** specifies, as the terminal end position of the straight line, the reference point the most distant from the moving body **702** at which the deviation from the straight line to the reference route satisfies a certain condition.

[0057] At **S903**, the control amount determination unit **304** specifies the most distant reference point at which deviation from a straight line indicating a current advancing direction of the moving body **100** is less than a threshold as the terminal end position of the straight line on the reference route. Processing of specifying the terminal end position of the straight line at this step will be described with reference to FIG. **8**.

[0058] The example illustrated in FIG. **8** illustrates an example of processing in which the moving body **702** that travels on the road **701** refers to the reference route **703** and determines the reference route length. In this example, the control amount determination unit **304** specifies the most distant reference point at which the deviation of the position from the straight line **802** indicating the current advancing direction of the moving body **702** is less than a threshold as the terminal end

position of the straight line. For example, the control amount determination unit **304** determines whether the dissociation degree between each reference point and the straight line **802** is less than the threshold. A triangle **803** represents a threshold range for determining the dissociation degree between each reference point and the straight line **802** on the basis of a distance from the standard reference point on the straight line **802** and a predetermined angle. The control amount determination unit **304** determines whether it is less than the threshold for each reference point while moving the standard reference point farther on the straight line **802** from a current position of the moving body **702**. At that time, some consecutive reference points may be sequentially evaluated. In a case where each reference point is within the threshold range illustrated by the triangle **803**, it is determined that it is less than the threshold for the reference point. For example, it is determined that the deviation of the position of the reference point **801** from the straight line **802** satisfies a certain condition (less than the threshold). Furthermore, it is assumed that a position of a next reference point of the reference point **801** deviates from the straight line **802** by the threshold or more. In this case, the control amount determination unit **304** specifies the position of the reference point **801** as the terminal end position of the straight line on the reference route [0059] At **S904**, the control amount determination unit **304** determines, as the reference route length, a length from the moving body **100** to the most distant terminal end position among the terminal end positions specified at **S901** to **S903**. Thereafter, the control amount determination unit **304** terminates the series of processing and returns to a caller.

[0060] Note that, in the above description, the example of determining the length of the reference route to be referred to for controlling the travel of the moving body **100** in the reference route length determination processing has been described. Note that, the present embodiment is not limited to the determination of the length of the reference route as long as a part of the reference route to be referred to for controlling the travel of the moving body **100** is determined. In the above-described embodiment, the case where the reference route is generated for the moving body to travel and is referred to has been described as an example. Note that, the reference route may be a route generated for other purposes as long as the reference route is referred to for controlling the travel of the moving body.

[0061] As described above, in the above-described embodiment, the reference route for the moving body **100** to travel is generated, and a part of the reference route to be referred to for controlling the travel of the moving body out of the reference route is determined. In particular, the control amount determination unit **304** identifies the terminal end of the straight line forming the reference route by first identification processing, and identifies the terminal end of the curved line forming the reference route or the change point of a predetermined curvature by second identification processing. Then, a part of the reference route is determined so as to include a position distant from the position of the moving body **100** out of the identified terminal ends. In this manner, it is possible to limit the reference route to a length of the route of a specific type such as a straight line or a curved line, and use a longer route that can be identified as the straight line or curved line. That is, it is possible to achieve both responsiveness to the route to travel and stability of the behavior.

[0062] Note that, the configuration of the above-described control unit **130** may function in various forms as the moving body control system. For example, the moving body control system may be configured in a form in which at least a part of the above-described control unit **130** is configured on a device outside the moving body **100**, for example, an external server. Alternatively, the moving body control system may be the moving body **100**, or may be incorporated in the moving body **100** (that is, may be the control unit **130**). A computer program for operating the above-described moving body **100** may be a computer program for causing one or more computers to function as each means of the moving body control system.

Summary of Embodiment

[0063] 1. A moving body control system that controls an operation of a moving body (for example, **100**) in above embodiment comprises: [0064] a route generation unit (for example, **303**) for

generating a reference route indicating a route from a position of the moving body on which the moving body travels; [0065] a part determination unit (for example, **304**) for determining a part of the reference route to be referred to for controlling travel of the moving body out of the reference route; and [0066] a control amount determination unit (for example, **304**) for determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, [0067] wherein [0068] the part determination unit determines a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing [0069] According to this embodiment, it is possible to achieve both responsiveness to the route to travel and stability of behavior.

[0070] 2. In the moving body control system according to the above-described embodiment, the first identification processing identifies the terminal end of the straight line forming the reference route on the basis of a coincidence degree between the reference route and a straight line from a specific point on the reference route to the moving body.

[0071] According to the embodiment, it becomes possible to separate a portion that can be approximated to a straight line starting from a moving body position out of the reference route as a straight line portion forming the reference route.

[0072] 3. In the moving body control system according to the above-described embodiment, the first identification processing identifies the terminal end of the straight line forming the reference route on the basis of deviation from a straight line indicating a current advancing direction of the moving body.

[0073] According to the embodiment, it becomes possible to separate a portion of the reference route extending in the advancing direction of the moving body out of the reference route as the straight line portion forming the reference route.

[0074] 4. In the moving body control system according to the above-described embodiment, the first identification processing performs both a first process based on a coincidence degree between the reference route and a straight line from a specific point on the reference route to the moving body, and a second process based on deviation from a straight line indicating a current advancing direction of the moving body, and identifies a terminal end distant from the moving body as a terminal end of the straight line.

[0075] According to the embodiment, it becomes possible to make a section that can be identified farther as a straight line from the position of the moving body a straight portion forming the reference route, and stabilize a behavior of the moving body by identifying a longer straight line portion.

[0076] 5. In the moving body control system according to the above-described embodiment, the second identification processing identifies a terminal end of a curved line forming the reference route on the basis of a change in curvature of a circular arc passing through a plurality of positions on the reference route.

[0077] According to the embodiment, it becomes possible to easily identify a terminal position of a curve portion of a road included in a range in which a curvature is constant out of the reference route and separate the curved line of the reference route as a curved line portion forming the reference route.

[0078] 6. In the moving body control system according to the above-described embodiment, a recognition unit (for example, **302**) recognizes a travel region for generating the reference route on the basis of an image obtained by capturing an external field of the moving body.

[0079] According to the embodiment, it becomes possible to identify a travel region of a road present in front of the moving body without additional information indicating a travel path such as a map, and autonomously travel the road.

[0080] The invention is not limited to the foregoing embodiments, and various variations/changes are possible within the spirit of the invention.

Claims

1. A moving body control system that controls an operation of a moving body, the moving body control system comprising: one or more processors; and a memory storing instructions which, when the instructions are executed by the one or more processors, cause the moving body control system to perform: generating a reference route indicating a route from a position of the moving body on which the moving body travels; determining a part of the reference route to be referred to for controlling travel of the moving body out of the reference route; and determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, wherein determining the part of the reference route includes determining a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.
2. The moving body control system according to claim 1, wherein the first identification processing includes identifying the terminal end of the straight line forming the reference route on the basis of a coincidence degree between the reference route and a straight line from a specific point on the reference route to the moving body.
3. The moving body control system according to claim 1, wherein the first identification processing includes identifying the terminal end of the straight line forming the reference route on the basis of deviation from a straight line indicating a current advancing direction of the moving body.
4. The moving body control system according to claim 1, wherein the first identification processing includes both a first process based on a coincidence degree between the reference route and a straight line from a specific point on the reference route to the moving body, and a second process based on deviation from a straight line indicating a current advancing direction of the moving body, and identifying a terminal end distant from the moving body as a terminal end of the straight line.
5. The moving body control system according to claim 1, wherein the second identification processing includes identifying a terminal end of a curved line forming the reference route on the basis of a change in curvature of a circular arc passing through a plurality of positions on the reference route.
6. The moving body control system according to claim 1, wherein the instructions further cause the moving body control system to perform recognizing a travel region for generating the reference route on the basis of an image obtained by capturing an external field of the moving body.
7. A moving body comprising: one or more processors; and a memory storing instructions which, when the instructions are executed by the one or more processors, cause the moving body control system to perform: generating a reference route indicating a route from a position of the moving body on which the moving body travels; determining a part of the reference route to be referred to for controlling travel of the moving body out of the reference route; and determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, wherein determining the part of the reference route includes determining a part of the reference route so as to include a terminal end distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.
8. A control method of a moving body control system that controls an operation of a moving body, the control method comprising: generating a reference route indicating a route from a position of the moving body on which the moving body travels; determining a part of the reference route to be

referred to for controlling travel of the moving body out of the reference route; and determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, wherein determining the part of the reference route includes determining a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.

9. A non-transitory computer-readable storage medium storing instructions for performing a control method of a moving body control system that controls an operation of a moving body, the control method comprising: generating a reference route indicating a route from a position of the moving body on which the moving body travels; determining a part of the reference route to be referred to for controlling travel of the moving body out of the reference route; and determining a control amount for controlling travel of the moving body with reference to the determined part of the reference route, wherein determining the part of the reference route includes determining a part of the reference route so as to include a position distant from the position of the moving body out of a terminal end of a straight line forming the reference route identified by first identification processing, and a terminal end or a change point of a predetermined curvature of a curved line forming the reference route identified by second identification processing.
